

TRIX: Training-Free Reranking for Instruction-Following Retrieval with Exclusions

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Abstract

Retrieval requests often combine a target topic with positive criteria, background context, and explicit exclusions. Frozen retrievers rely heavily on lexical and semantic entity matching, overlooking how these instructions define relevance. This failure is acute for exclusions, where an excluded entity can become a strong matching signal. Existing remedies often require instruction training or representation adaptation. We introduce TRIX, a training-free reranking framework for frozen retrievers. TRIX preserves the complete request for retrieval and decomposes it into affirmative and exclusion views, focusing positive evidence and removing irrelevant background context. The two views can still share topic information, causing raw exclusion scores to penalize relevant content. TRIX removes the exclusion similarity predicted by affirmative relevance, then uses the residual to reduce positive scores and suppress exclusion matches. It requires no model training, retriever updates, or document re-encoding. Across three benchmarks and six frozen retrievers, TRIX improves retrieval effectiveness and instruction sensitivity, increasing the latter by up to 16.4 points and outperforming a matched training-free baseline on NegConstraint by up to 3.4 points.

1 Introduction

Retrieval requests often combine a target topic with positive relevance criteria, background context, and explicit exclusions. Frozen retrievers rely mainly on lexical and semantic matching, but these signals do not reliably capture how each part of an instruction affects relevance. The problem is especially clear for exclusions because the excluded entity remains part of the request. In lexical retrieval, it contributes matching terms like any other query content. Dense retrievers face a related limitation: pooled embeddings compress topical content and compositional roles into a single vector, and cosine scoring can keep texts with similar content but different negation or scope close in the embedding space (Ralev et al. 2026). A document can therefore receive a higher score for matching an entity that the user explicitly asks to exclude. Figure 1 illustrates a request for evidence that a US college education improves employment or income while excluding documents about Manhattan colleges

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such as Columbia University. Documents about employment outcomes for Columbia graduates match the topic and entities in the request, yet they violate the exclusion. This failure is consistent with prior findings that neural rankers often fail to adjust their rankings when negation changes the relevance conditions (Weller, Lawrie, and Durme 2024; Weller et al. 2025a).

Separating the instruction into positive and exclusion views is a natural way to make these roles explicit. However, the two views can still share topic information. In Figure 1, the fifth-ranked document satisfies the request and does not discuss the excluded Manhattan colleges, but retrieval scores are based mainly on lexical or embedding similarity. Its broader discussion of US colleges and graduate outcomes therefore raises its scores under both the affirmative and exclusion views. Directly subtracting the exclusion score would penalize a relevant document. An effective correction must distinguish evidence of the excluded content from similarity caused by the shared topic.

Prior methods introduce instruction control through instruction-trained retrieval, inference-time query adaptation, or logic-based constraint modeling. Instruction-trained retrievers such as Promptriever (Weller et al. 2025b), Dual-View Training (Zeng et al. 2026), and InstEmb (Gao et al. 2026) learn instruction-conditioned representations, but require additional supervision and model training. Inference-time query adaptation methods such as DEO (Lee et al. 2026) modify the query embedding at inference time, but direct optimization can move the query vector away from the embedding structure learned by the retriever, making its similarity to document embeddings less reliable. Logic-based constraint methods such as NS-IR (Xu et al. 2025) and OrLog (Hoveyda et al. 2026) translate instructions into logical structures and apply symbolic or probabilistic inference over the retrieved candidates. The additional translation, candidate-level model calls, and formal inference increase online retrieval latency.

We therefore introduce TRIX, a training-free reranking framework that corrects the scores of a frozen retriever without updating its representations, re-encoding documents, or adding candidate-level constraint inference. TRIX retains the complete instruction for initial retrieval, derives an affirmative view that focuses the positive criteria and removes irrelevant background context, and represents excluded content in a separate view. It uses a query-local residual as a proxy

for exclusion evidence beyond topical match. Based on this signal, the final scoring rule preserves or reduces affirmative support and applies an exclusion penalty.

Across an instruction-following benchmark (FollowIR) and two exclusion-aware benchmarks (NegConstraint and ExcluIR), TRIX improves instruction-aware ranking across sparse and dense frozen retrievers. It raises instruction sensitivity by up to 16.4 points and outperforms a matched training-free baseline on NegConstraint by up to 3.4 points.

Our contributions are:

- We introduce a training-free reranking framework that uses full, affirmative, and exclusion views to handle positive criteria, background context, and exclusions without retriever updates or document re-encoding.
- We use residual correction to obtain a query-local proxy for exclusion evidence beyond topical match, then combine relevance promotion and exclusion suppression according to this signal.
- Across three benchmarks and multiple frozen retrievers, we demonstrate consistent improvements and isolate the effects of view construction, residual scoring, and final score correction, while maintaining general retrieval effectiveness.

2 Related Work

Prior work on instruction control in retrieval falls into three groups: instruction-trained retrieval, inference-time query adaptation, and logic-based constraint modeling.

2.1 Instruction-Trained Retrieval

Instruction-trained retrievers learn to score documents according to natural-language instructions. TART uses multi-task instruction tuning with natural-language task descriptions (Asai et al. 2023). INSTRUCTOR learns a shared instruction-conditioned embedding space (Su et al. 2023). HypeR composes task-dependent prompts for cross-domain retrieval (Cai et al. 2023). More recent systems train directly with relevance instructions, including FollowIR-7B (Weller et al. 2025a), Promptriever (Weller et al. 2025b), and InF-IR (Zhuang et al. 2025). Dual-View Training adds polarity-reversed supervision (Zeng et al. 2026), while InstEmb combines input-intrinsic and output-aware signals when learning instruction-following embeddings (Gao et al. 2026). CoDeR learns constraint compatibility through polarity supervision, requiring an additional trained encoder and corpus re-encoding (Yin, Tang, and Du 2026).

These training-based methods require instruction-specific data and updates to the retriever parameters. Targeted negation augmentation likewise requires additional training data (Singh, Kumar, and Sridhar 2023), and supporting new instruction types can require further data and retraining. TRIX keeps the retriever and its representation function fixed and introduces instruction control during reranking.

2.2 Inference-Time Query Adaptation

Inference-time methods modify the query text or representation without retraining the retriever (Ma et al. 2023). HyDE

generates a hypothetical document to guide retrieval (Gao et al. 2023). Query2Doc expands the query with a generated pseudo-document (Wang, Yang, and Wei 2023). RAG-Fusion combines rankings from multiple generated queries (Rackauckas 2024). Stage-Aware Decomposition retains the complete query for initial retrieval and additively aggregates decomposed condition scores during reranking (Yin et al. 2026). These methods add or reformulate topical content but do not handle exclusion constraints. Excluded entities can therefore remain strong lexical or semantic matching signals.

Other methods directly modify or construct query representations. DEO derives positive and negative subqueries and optimizes the query embedding toward the former and away from the latter (Lee et al. 2026). Because the resulting vector is not generated by the retriever’s query encoder, it may deviate from the learned embedding structure and make its similarity to document embeddings less reliable. In vision-language retrieval, SpaceVLM-DRC uses angular composition to construct three negation-aware query directions and combines their image similarities with a direct image-space repulsion term (Pokhrel and Sheshappanavar 2026). This design assumes a unit-normalized shared image-text embedding space and treats direct similarity to the negated concept as evidence that an image contains it. Text retrieval therefore requires a scoring rule that supports sparse retrievers and distinguishes exclusion evidence from similarity caused by shared topic information.

TRIX formulates instruction control as score correction over the retrieved documents. It retains the complete query for candidate retrieval, keeps the query and document representations unchanged, estimates exclusion evidence from the relation between affirmative and exclusion scores, and combines relevance promotion with exclusion suppression. This score-based formulation applies to sparse and dense text retrievers.

2.3 Logic-Based Constraint Modeling

Logic-based methods represent constraints explicitly before reranking. NS-IR translates queries and documents into first-order logic and combines semantic alignment with connective constraints for negative-constraint retrieval (Xu et al. 2025). OrLog decomposes entity-seeking queries into atomic predicates and a logical form, then uses predicate plausibility scores in probabilistic logic inference (Hoveyda et al. 2026).

These methods enforce constraint composition through formal representations and candidate-level inference. The required logical translation, model calls, and formal inference increase online latency and make retrieval depend on the quality of the intermediate logical representation. TRIX operates directly on the scores of a frozen retriever, without translating documents into logical forms or invoking an external inference engine.

3 Methodology

Figure 1 presents the three stages of TRIX. Section 3.1 constructs full, affirmative, and exclusion evidence views. Section 3.2 computes a residual score as a query-local proxy for exclusion evidence beyond topical match. Section 3.3

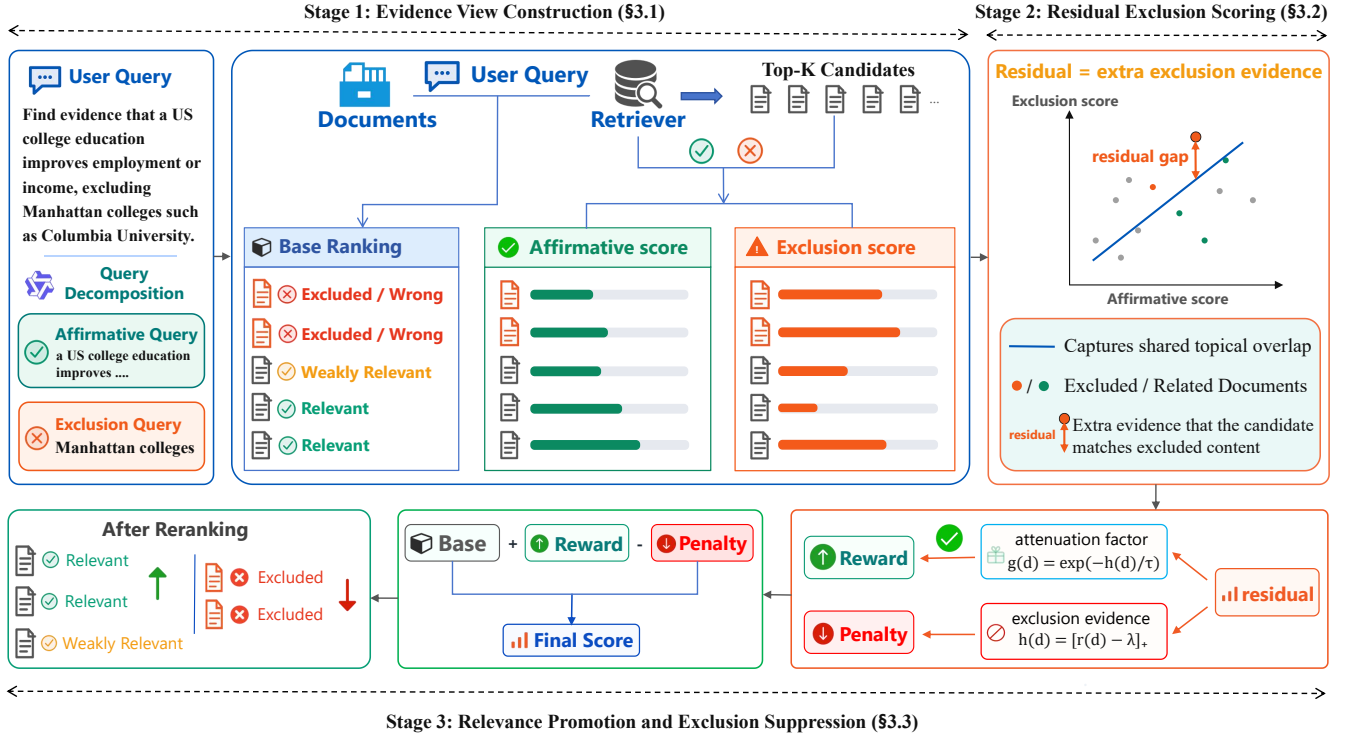


Figure 1: Overview of TRIX. Stage 1, Evidence View Construction (Section 3.1), constructs full, affirmative, and exclusion views and retrieves candidates with the full view. Stage 2, Residual Exclusion Scoring (Section 3.2), forms a query-local residual after removing shared score variation over the retrieved documents. Stage 3, Relevance Promotion and Exclusion Suppression (Section 3.3), converts this evidence into the final ranking correction.

uses this score to promote affirmative relevance and suppress strong exclusion matches.

3.1 Evidence View Construction

Stage 1 in Figure 1 constructs full, affirmative, and exclusion views while retaining the complete request for initial retrieval. The base ranking shows why the auxiliary views are needed: two documents about excluded Manhattan colleges are ranked first because they match both the desired college-employment topic and the excluded entities named in the request. The retriever treats both forms of lexical or embedding similarity as positive signals without reliably distinguishing their roles in the instruction.

TRIX receives a natural-language request u containing the retrieval topic and its instructions. The frozen retriever uses the complete request as the full query $Q_{\text{full}} = u$ and returns the top- K candidate set $\mathcal{C}_u$. An LLM decomposes u once into two auxiliary queries: Q_{pos} states the content that supports relevance, and Q_{neg} states the content to exclude. If the request contains no exclusion, the decomposer returns $Q_{\text{neg}} = [\text{NONE}]$. The candidate set is always retrieved with Q_{full} ; the auxiliary views are used only for reranking.

Table 1 illustrates this decomposition using the running example. The affirmative view retains the desired content, and the exclusion view converts the rejected content into a directly scorable description.

View	Example text
Q_{full}	Find evidence that a US college education improves employment prospects or income. Exclude documents about colleges in Manhattan, such as Columbia University.
Q_{pos}	Evidence that a US college education improves employment prospects or income.
Q_{neg}	Colleges in Manhattan, such as Columbia University.

Table 1: Evidence views for the running example in Figure 1. The affirmative view states the desired evidence, while the exclusion view states the content to be excluded.

The three views serve different roles. The full query retains the complete topic, context, and instruction during candidate generation. The affirmative view keeps the target topic and positive relevance requirements while removing the exclusion clause and background information that does not determine relevance. The exclusion view states the rejected content as a positive description, allowing the retriever to score whether a document matches that content. In addition to supporting relevance, the affirmative score provides a reference for the topical component shared with the exclusion score. All views use the same frozen scorer.

Let $s(Q, d)$ denote the score assigned to document d under query Q . For each $d \in \mathcal{C}_u$, the three channels are

$$\begin{aligned} S_{\text{full}}(d) &= s(Q_{\text{full}}, d), \\ S_{\text{pos}}(d) &= s(Q_{\text{pos}}, d), \\ S_{\text{neg}}(d) &= s(Q_{\text{neg}}, d). \end{aligned} \quad (1)$$

For a dense retriever, $s(Q_x, d) = \cos(f_q(Q_x), f_d(d))$ is the embedding similarity; for BM25, it is the lexical retrieval score. Dense document embeddings are computed once and reused across views, while BM25 uses one fixed index.

3.2 Residual Exclusion Scoring

Stage 2 in Figure 1 uses the residual as a query-local proxy for exclusion evidence beyond topical match. This avoids strongly penalizing candidates whose exclusion scores are high only because they closely match the topic. In the score plot, the fitted line represents the exclusion score expected from a candidate’s affirmative score. The fifth, relevant candidate has high scores in both views but lies close to the fitted line, indicating that its exclusion score is expected from the shared college background. The highlighted orange candidate lies farther above the line and therefore produces a larger positive residual, indicating additional evidence that it matches the excluded Manhattan colleges.

TRIX addresses this ambiguity by estimating how much exclusion similarity is expected from the affirmative score and using the deviation from this relation as an exclusion proxy. Because all candidates in $\mathcal{C}_u$ are retrieved for the same request, the two scores can rise together when both respond to the shared topic. An affine fit provides a simple per-request estimate of this score dependence. The intercept captures a common offset, and the slope captures how the exclusion score changes with the affirmative score. If the two score channels have no systematic relation, the estimated slope is expected to be near zero. The fitted value then becomes approximately constant, so the standardized residual reduces to the normalized exclusion score up to centering and scaling.

TRIX first robustly standardizes all three score channels over the retrieved set $\mathcal{C}_u$:

$$z_x(d) = \frac{S_x(d) - \text{median}_{d' \in \mathcal{C}_u} S_x(d')}{\text{MAD}_{d' \in \mathcal{C}_u}(S_x(d')) + \epsilon}, \quad (2)$$

$x \in \{\text{full}, \text{pos}, \text{neg}\},$

where $\text{MAD}(S) = \text{median}|S - \text{median}(S)|$ and $\epsilon > 0$ prevents division by zero. This normalization aligns query-specific offsets and scales while limiting the influence of extreme scores (Rousseeuw and Croux 1993). TRIX then fits the affine relation with Huber regression:

$$(\hat{a}_u, \hat{b}_u) = \arg \min_{a, b} \sum_{d \in \mathcal{C}_u} \rho_\delta(z_{\text{neg}}(d) - a - bz_{\text{pos}}(d)), \quad (3)$$

where ρ_δ is the Huber loss with transition point δ . The fit captures the dominant score association, while the Huber loss limits the influence of large deviations (Huber 1964). It provides a query-specific reference for exclusion similarity. If the unnormalized positive-score channel has $\text{MAD}(S_{\text{pos}}) < \epsilon$, we set $\hat{b}_u = 0$ and use a constant fitted

value. For $e(d) = z_{\text{neg}}(d) - \hat{a}_u - \hat{b}_u z_{\text{pos}}(d)$, we robustly standardize the residual:

$$r(d) = \frac{e(d) - \text{median}_{d' \in \mathcal{C}_u} e(d')}{\text{MAD}_{d' \in \mathcal{C}_u}(e(d')) + \epsilon}. \quad (4)$$

The second standardization makes residual magnitudes comparable across requests: $r(d) = 0$ is the median fitted deviation, and one unit is measured relative to the within-request residual dispersion. This common scale allows the same decision parameters to be used across queries and retrievers. Positive $r(d)$ indicates more exclusion similarity than predicted by the document’s affirmative score; negative $r(d)$ indicates less. The residual is a relative score-space proxy, not a calibrated probability of constraint violation. As detailed in the next stage, a residual value above the shared boundary reduces the affirmative reward and activates an exclusion penalty. Numerical details are given in Appendix E.

3.3 Relevance Promotion and Exclusion Suppression

Stage 3 in Figure 1 uses the residual proxy to control the final score correction. Candidates with strong affirmative support and a small residual retain a positive correction, while candidates with a large positive residual lose this support and receive a penalty. This raises relevant candidates and pushes the two initially top-ranked but excluded documents downward. We implement this correction by first retaining the positive part of the normalized affirmative score:

$$p(d) = [z_{\text{pos}}(d)]_+. \quad (5)$$

Because $z_{\text{pos}}(d)$ is median-centered, $p(d)$ rewards only documents whose affirmative score is above the candidate-set median. We activate the exclusion correction beyond a shared boundary $\lambda \geq 0$:

$$h(d) = [r(d) - \lambda]_+, \quad g(d) = \exp\left(-\frac{h(d)}{\tau}\right), \quad \tau > 0, \quad (6)$$

where $h(d)$ is the positive residual excess above the boundary and $g(d) \in (0, 1]$ is the reward attenuation factor. Below the boundary, $h(d) = 0$ and $g(d) = 1$; above it, $h(d)$ grows linearly while $g(d)$ decreases smoothly. The final score is

$$S_{\text{final}}(d) = z_{\text{full}}(d) + p(d)g(d) - h(d). \quad (7)$$

The term $p(d)g(d)$ retains the affirmative reward below the residual boundary and attenuates it once the boundary is crossed. The term $-h(d)$ directly suppresses candidates with a large positive residual, so the correction can lower a document below its full-query score. Because the three score components are derived from robustly standardized scores, we combine them using fixed unit weights. These weights remain unchanged across datasets and retrievers. For fixed $z_{\text{full}}(d)$ and $p(d)$, Eq. 7 is therefore non-increasing in $r(d)$. When no exclusion view is present, TRIX sets $h(d) = 0$ and $g(d) = 1$, reducing the score to $z_{\text{full}}(d) + p(d)$.

4 Experiments

4.1 Experimental Setup

Benchmarks and Metrics. FollowIR (Weller et al. 2025a) measures instruction-following retrieval on Core17, Ro-

bust04, and News21. We report aggregate retrieval effectiveness (Score) and p-MRR for instruction sensitivity; positive p-MRR indicates that rank changes follow the relevance instruction. NegConstraint (Xu et al. 2025) and ExcluIR (Zhang et al. 2025) evaluate exclusion-aware corpus ranking (MAP and nDCG@10) and desired-excluded pair ordering (MRR and Right Rank), respectively. Eight BEIR datasets (Thakur et al. 2021) measure general retrieval performance using nDCG@10 and MAP@100.

Models and Settings. Frozen Qwen3-4B (Yang et al. 2025) produces both auxiliary views in one generation. All 3,858 decompositions are structurally valid; a stratified manual audit of 90 requests finds that the affirmative view preserves the topic and positive criteria in every case and the exclusion view covers 67 of 69 applicable exclusions, with two cases judged unclear (Appendix F). FollowIR uses BM25 (Robertson and Zaragoza 2009) together with the frozen E5-Mistral (Wang et al. 2024), BGE-large (Xiao et al. 2024), and RepLLaMA (Ma et al. 2024); NegConstraint and ExcluIR use BGE-small-en-v1.5, BGE-large-en-v1.5, BGE-M3 (Xiao et al. 2024), and RepLLaMA. We rerank 1,000 full-query candidates except on ExcluIR, where the benchmark protocol requires full-corpus scoring. Matched variants share candidates, views, templates, and document representations. We select $\lambda=0.5$ and $\tau=0.1$ once on the training corpus released with FollowIR using RepLLaMA, then freeze them across all evaluation datasets and retrievers; no labels from the reported evaluation sets are used. Detailed protocols and configurations are provided in Appendices C and E. We test matched differences using a two-sided paired randomization test with Holm correction and a significance threshold of $p < 0.05$.

Baselines and Controlled Comparisons. On FollowIR, FollowIR-7B (Weller et al. 2025a), Promptriever (Weller et al. 2025b), and InstEmb (Gao et al. 2026) are included as instruction-trained baselines. We also reproduce HyDE (Gao et al. 2023), Query2Doc (Wang, Yang, and Wei 2023), and ConvSearch-R1 (Jiang et al. 2025) with RepLLaMA. On the exclusion benchmarks, NS-IR (Xu et al. 2025), DEO (Lee et al. 2026), and SEAL (Zhang et al. 2025) are included as prior methods. We reproduce DEO using the same BGE-large top-1,000 candidates, benchmark split, and evaluation implementation as TRIX on NegConstraint; published-reference rows are reported separately from the matched Base/+TRIX comparisons.

4.2 Main Results on Instruction-Following Retrieval

Table 2 reports aggregate FollowIR results. The lower block contains matched Base/+TRIX pairs spanning BM25 and three dense retriever families; the upper block reports instruction-trained retrievers. Per-dataset results are reported in Appendix A.

TRIX improves both retrieval effectiveness and instruction sensitivity for every frozen backbone, spanning BM25 and three dense retrievers. Before reranking, the three dense backbones have negative average p-MRR, indicating that their rankings often fail to follow the relevance instruction. TRIX

Method	Score $\uparrow$	p-MRR $\uparrow$
<i>Instruction-trained retrievers</i>		
FollowIR-7B	24.8	+12.2
Promptriever	26.1	+11.2
InstEmb (Llama-3-8B)	28.5	+15.6
<i>Matched frozen-backbone comparisons</i>		
BM25	15.2	+1.1
+TRIX	17.3	+17.5
E5-Mistral	26.1	−2.7
+TRIX	28.0	+12.1
BGE-large	19.2	−2.2
+TRIX	23.3	+6.5
RepLLaMA	23.8	−3.3
+TRIX	28.7	+12.3

Table 2: Aggregate FollowIR results. Score averages MAP on Core17 and Robust04 with nDCG on News21. Each +TRIX row is matched to the backbone above it; values are percentages ($\times 100$). Best results within each comparable block are shown in bold.

Method	Score $\uparrow$	p-MRR $\uparrow$
RepLLaMA	23.8	−3.3
HyDE	23.3	+1.4
Query2Doc	25.6	−1.1
ConvSearch-R1	23.4	+0.2
TRIX	28.7	+12.3

Table 3: Aggregate FollowIR comparison with query rewriting and expansion using RepLLaMA. Score averages MAP on Core17 and Robust04 with nDCG on News21; values are percentages ($\times 100$). Best results are shown in bold.

makes p-MRR positive while also raising Score; with RepLLaMA, Score increases from 23.8 to 28.7 and p-MRR from −3.3 to +12.3; both improvements are statistically significant ($p < 0.05$). Each Base/+TRIX pair ranks the same candidates with the same document representations; TRIX only corrects their scores.

Comparison with Query Rewriting and Expansion. Table 3 compares TRIX with representative inference-time rewriting and expansion methods using RepLLaMA. Each method uses its original query-generation and retrieval procedure. Results for Core17, Robust04, and News21 are provided in Appendix Table 8.

TRIX outperforms all compared rewriting and expansion methods on both aggregate metrics, reaching 28.7 Score and +12.3 p-MRR. Existing rewriting methods are mainly designed to make the target topic easier to match. For example, HyDE generates a hypothetical relevant document, while Query2Doc expands the query with a generated pseudo-document. Both add positive topical context, but neither represents excluded content as evidence that should lower a document’s rank. Excluded entities can therefore remain strong lexical or semantic matching signals after rewriting. TRIX separates affirmative and exclusion evidence before

Backbone / Method	NegConstraint		ExcluIR	
	MAP $\uparrow$	nDCG@10 $\uparrow$	MRR $\uparrow$	RR $\uparrow$
<i>External and reproduced baselines</i>				
NS-IR (GPT-4o)	53.3	56.5	—	—
DEO (BGE-large)	73.2	78.8	74.1	79.3
SEAL (trained)	—	—	78.4	78.0
<i>Matched frozen-backbone comparisons</i>				
BGE-small	68.4	75.2	75.0	60.1
+TRIX	76.9	82.3	77.1	81.3
BGE-large	63.8	71.9	73.4	56.0
+TRIX	76.6	81.8	78.2	82.2
BGE-M3	64.3	74.1	78.9	67.0
+TRIX	74.7	81.1	79.7	83.7
RepLLaMA	72.8	79.4	75.9	62.1
+TRIX	80.0	84.8	87.2	89.9

Table 4: Results on NegConstraint and ExcluIR. NegConstraint uses top-1,000 candidates, whereas ExcluIR scores its full 90,406-document corpus. Each +TRIX row uses the same candidate pool and document representations as its base. DEO results are our BGE-large reproductions; its NegConstraint run uses the same candidate and evaluation setup as the BGE-large TRIX experiment. Values are percentages ($\times 100$). Best results within each comparable block are shown in bold.

correcting the ranking scores.

4.3 Evaluation on Exclusion-Aware Benchmarks

We evaluate TRIX on two benchmarks designed for retrieval with negative constraints. NegConstraint (Xu et al. 2025) evaluates compositional constraints over a ranked corpus, while ExcluIR (Zhang et al. 2025) compares desired and excluded documents in pairs.

NegConstraint reranks 1,000 candidates, whereas ExcluIR scores its full corpus; Appendix C provides the evaluation protocol.

TRIX improves both metrics for every matched backbone on both benchmarks. RepLLaMA provides the strongest overall results: TRIX raises its MAP from 72.8 to 80.0 and nDCG@10 from 79.4 to 84.8 on NegConstraint, while improving MRR from 75.9 to 87.2 and RR from 62.1 to 89.9 on ExcluIR. For RepLLaMA, the improvements in both metrics are statistically significant on each benchmark ($p < 0.05$). Under the matched BGE-large NegConstraint setup, TRIX reaches 76.6 MAP and 81.8 nDCG@10, exceeding DEO by 3.4 and 3.0 points, respectively. It also exceeds the reproduced DEO results and the reported SEAL results on ExcluIR.

4.4 General Retrieval Performance

Table 5 reports average results on eight BEIR datasets; the per-dataset breakdown is in Appendix G.

Average nDCG@10 changes from 60.02 to 60.23 and MAP@100 from 44.97 to 44.92. These small aggregate changes show that TRIX improves instruction sensitive re-

Method	Avg. nDCG@10 $\uparrow$	Avg. MAP@100 $\uparrow$
BGE-large	60.02	44.97
+TRIX	60.23	44.92

Table 5: General retrieval performance on eight BEIR datasets with BGE-large. TRIX maintains average retrieval effectiveness, with changes of +0.21 nDCG@10 and -0.05 MAP@100; values are percentages ($\times 100$).

Variant	Decision score	Score	p-MRR
<i>Evidence construction</i>			
Full-query baseline	z_{full}	23.8	-3.3
Positive-view scoring	$z_{\text{full}} + p$	25.5	-0.7
Exclusion-view subtraction	$z_{\text{full}} - z_{\text{neg}}$	25.8	$+7.6$
<i>Exclusion-signal construction</i>			
Raw exclusion subtraction	$z_{\text{full}} + p - z_{\text{neg}}$	24.0	$+14.9$
Residual exclusion subtraction	$z_{\text{full}} + p - r$	26.6	$+12.7$
<i>TRIX score composition</i>			
TRIX w/o reward attenuation	$z_{\text{full}} + p - h$	27.1	$+10.8$
TRIX w/o exclusion penalty	$z_{\text{full}} + pg$	26.2	$+8.9$
Full TRIX	$z_{\text{full}} + pg - h$	28.7	+12.3

Table 6: Controlled mechanism ablation on FollowIR. The three blocks isolate evidence construction, exclusion-signal construction, and the two operations in TRIX score composition. Score averages MAP on Core17 and Robust04 with nDCG on News21; all values are percentages ($\times 100$). Results for the full system are shown in bold.

trieval while maintaining strong general retrieval performance.

4.5 Ablation Study

Table 6 organizes the ablation around three design steps: evidence view construction, residual exclusion scoring, and final score correction. Here p , r , h , and g denote the positive score, residual exclusion score, thresholded exclusion penalty, and reward attenuation factor defined in Section 3. All variants use the same candidates, generated views, frozen encoder, templates, embeddings, and hyperparameters.

Evidence Views. The focused positive view raises Score from 23.8 to 25.5 and p-MRR from -3.3 to -0.7 , confirming that it contributes evidence beyond the full instruction. The exclusion view also contributes independently: subtracting its score raises Score to 25.8 and p-MRR to $+7.6$. The two views therefore provide complementary signals for relevance promotion and exclusion suppression.

Residual Exclusion Evidence. Raw subtraction aggressively lowers documents with high exclusion scores, including both exclusion matches and relevant documents whose scores are high because of shared topic content. The resulting rank changes produce a higher p-MRR of $+14.9$, but the collateral suppression of relevant documents lowers Score to 24.0. Measuring exclusion evidence relative to the affirmative score makes the correction more selective: p-MRR

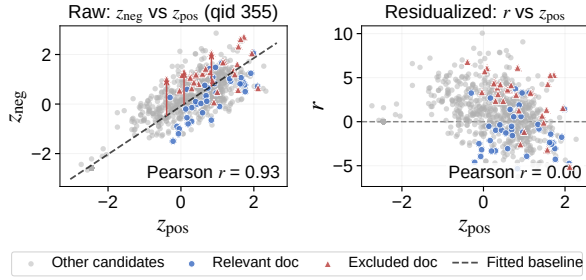


Figure 2: Residualization for an illustrative query. Blue circles denote relevant documents, red triangles denote excluded documents, and gray circles denote other candidates. Left: the dashed Huber fit captures the strong association between affirmative and exclusion scores. Right: standardized deviations from the fit; the dashed horizontal line marks $r = 0$.

decreases to +12.7, while Score rises to 26.6. The residual therefore reduces broad rank disruption and preserves more relevant documents.

Score Correction. Full TRIX improves over the variant without reward attenuation by 1.6 Score and 1.5 p-MRR points, showing that strong residual evidence should also reduce the positive contribution. Adding the explicit penalty to the attenuation-only variant yields gains of 2.5 Score and 3.4 p-MRR points. Together, the two comparisons support the asymmetric combination of attenuating affirmative support and directly suppressing strong exclusion matches.

4.6 Mechanism Analysis

Residual Signal. Figure 2 illustrates how residualization changes the meaning of the exclusion score. In the left panel, z_{neg} increases strongly with z_{pos} (Pearson $r = 0.93$), showing that the two views share substantial topical information. The dashed Huber fit estimates the exclusion score expected from this association. Many blue relevant documents have relatively high raw exclusion scores, yet lie below the fitted relation; directly subtracting z_{neg} would therefore penalize them for similarity already explained by their affirmative scores.

The right panel shows the standardized residual after removing the fitted component. Its correlation with z_{pos} falls to 0.00, indicating that the shared score trend has been removed. Many blue relevant documents now fall below $r = 0$ and therefore remain below the correction boundary, preserving their affirmative reward and avoiding an exclusion penalty. Most red excluded documents retain positive residuals because their exclusion similarity exceeds the level predicted by their affirmative score. This more selective correction matches Table 6: compared with raw subtraction, residualization reduces p-MRR from +14.9 to +12.7 as the rank changes become less aggressive, while increasing Score from 24.0 to 26.6 by preserving more relevant documents.

Ranking Behavior. Figure 3 visualizes the document-level rank changes produced by TRIX.

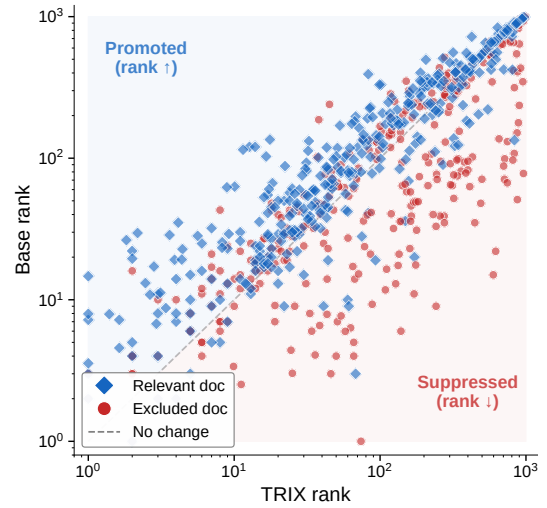


Figure 3: Rank changes produced by TRIX on FollowIR exclusion queries. The vertical axis gives the initial rank from the frozen retriever, and the horizontal axis gives the rank after TRIX adds the affirmative reward and subtracts the exclusion penalty. Points above the diagonal are promoted, points below it are suppressed, and points on the diagonal are unchanged. Blue diamonds denote relevant documents, while red circles denote documents that violate the exclusion constraint.

Because smaller rank values are better, a point above the diagonal has a better TRIX rank than its base rank, whereas a point below the diagonal has been pushed downward. The blue diamonds concentrate above the diagonal, showing that most relevant documents receive a net promotion from the affirmative reward. The red circles concentrate below the diagonal, showing that documents violating the exclusion are generally suppressed by the residual-controlled penalty. Points close to the diagonal receive smaller corrections and retain similar ranks. This separation shows that TRIX directs its two scoring operations toward the intended documents: relevance promotion primarily benefits relevant documents, and exclusion suppression primarily lowers documents that violate the constraint.

5 Conclusion and Limitations

This paper presented TRIX, a training-free score correction method for frozen retrievers. Its evidence views improve relevance scoring, while residual exclusion scoring reduces penalties caused by shared topic information. Across sparse and dense retrievers, the resulting correction improves instruction-sensitive ranking while maintaining general retrieval performance.

TRIX requires useful decomposed views and adequate initial candidate coverage; as a reranker, it cannot recover documents absent from the retrieved set. The residual may also be less informative when the exclusion view provides little evidence in that set. Stronger decomposition and candidate generation are natural directions for future work.

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A Full FollowIR Results

Table 7 provides the per-dataset results underlying the aggregate comparison in Table 2. Results for instruction-trained methods are taken from their source papers (Weller et al. 2025a,b; Gao et al. 2026).

(a) Retrieval effectiveness $\uparrow$				
Method	Core17	Robust04	News21	Score
<i>Instruction-trained retrievers</i>				
FollowIR-7B	20.0	24.8	29.6	24.8
Promptriever	21.6	28.3	28.5	26.1
InstEmb (Llama-3-8B)	24.0	29.2	32.3	28.5
<i>Matched frozen-backbone comparisons</i>				
BM25	10.8	13.8	21.0	15.2
+TRIX	13.2	14.0	24.8	17.3
E5-Mistral	20.4	27.9	30.1	26.1
+TRIX	24.4	26.7	32.8	28.0
BGE-large	16.8	19.4	21.4	19.2
+TRIX	22.1	21.9	26.0	23.3
RepLLaMA	23.2	25.7	22.5	23.8
+TRIX	25.9	28.2	32.1	28.7
(b) Instruction sensitivity (p-MRR $\uparrow$)				
Method	Core17	Robust04	News21	Average
<i>Instruction-trained retrievers</i>				
FollowIR-7B	+16.5	+13.7	+6.3	+12.2
Promptriever	+15.4	+11.7	+6.4	+11.2
InstEmb (Llama-3-8B)	+22.4	+19.1	+5.4	+15.6
<i>Matched frozen-backbone comparisons</i>				
BM25	+3.7	−0.5	0.0	+1.1
+TRIX	+18.4	+13.8	+20.3	+17.5
E5-Mistral	−0.3	−6.9	−0.9	−2.7
+TRIX	+5.7	+19.6	+11.1	+12.1
BGE-large	−2.0	−5.2	+0.5	−2.2
+TRIX	+4.3	+9.0	+6.3	+6.5
RepLLaMA	+1.1	−9.1	−1.8	−3.3
+TRIX	+16.2	+14.1	+6.7	+12.3

Table 7: Per-dataset FollowIR results. The two panels report retrieval effectiveness and p-MRR for the same systems; values are percentages ($\times 100$).

B Additional Ablations and Sensitivity

The main text isolates the positive view, residual exclusion signal, reward attenuation, and explicit penalty. The following results compare alternative residual estimators and examine sensitivity to the shared configuration.

B.1 Estimator and Normalization Choices

Table 9 compares the robust estimator, normalization, and residual construction used to obtain r .

Table 10 varies one factor at a time around the main RepLLaMA configuration. The $K = 1000$ row reproduces the main FollowIR result (28.7 Score and +12.3 p-MRR), resolving the candidate-depth setting used by Tables 2 and 3. All tested depths improve over the RepLLaMA base result of 23.8 Score and −3.3 p-MRR. From $K = 10$ to

(a) Retrieval effectiveness $\uparrow$

Method	Core17	Robust04	News21	Score
RepLLaMA	23.2	25.7	22.5	23.8
HyDE	23.6	24.9	21.4	23.3
Query2Doc	25.9	26.1	24.9	25.6
ConvSearch-R1	22.3	25.9	22.0	23.4
TRIX	25.9	28.2	32.1	28.7

(b) Instruction sensitivity (p-MRR $\uparrow$)

Method	Core17	Robust04	News21	Average
RepLLaMA	+1.1	−9.1	−1.8	−3.3
HyDE	+6.5	−2.9	+0.7	+1.4
Query2Doc	+8.0	−7.5	−3.8	−1.1
ConvSearch-R1	+3.0	−2.5	+0.2	+0.2
TRIX	+16.2	+14.1	+6.7	+12.3

Table 8: Per-dataset FollowIR comparison with query rewriting and expansion using RepLLaMA. The two panels report retrieval effectiveness and p-MRR; values are percentages ($\times 100$).

Variant	Modification	Score	p-MRR
Full TRIX	None	28.7	+12.3
TRIX w/ OLS	Replace Huber regression	26.9	+10.6
TRIX w/ mean/std scaling	Replace median/MAD	26.8	+8.5
TRIX w/o residual centering	Omit residual recentering	27.1	+10.8

Table 9: Secondary TRIX design choices on FollowIR. Each row changes one estimation or normalization choice while retaining the controlled candidate set and three evidence views.

$K = 200$, p-MRR increases steadily from +16.1 to +19.8, while Score rises from 26.3 to 27.9 at $K = 100$ before decreasing to 27.3 at $K = 200$. Expanding the depth to the default $K = 1000$ yields the highest Score of 28.7 but lowers p-MRR to +12.3. The updated results therefore show a depth-dependent trade-off: shallower candidate sets produce larger measured instruction-sensitive rank changes, whereas the full depth provides the best aggregate retrieval effectiveness. Because the same K determines retrieval coverage, the pool used for robust normalization and Huber fitting, and the documents reranked before evaluation, this analysis measures their joint effect. Separating candidate coverage from score-estimation effects would require holding retrieval depth fixed while varying only the estimation pool. In the one-factor parameter sweeps, increasing either λ or τ reduces both reported metrics relative to the tested defaults.

Instruction decomposition averages 2.81 seconds per query (standard deviation 1.20, range 1.51–6.80 seconds), corresponding to a throughput of 0.36 decomposition queries per second. After the three retrieval-score channels are available, score normalization, Huber fitting, and final reranking take 0.4 ms per query. This post-scoring measurement excludes query decomposition, query encoding, and candidate retrieval, and therefore is not an end-to-end latency

Analysis	Setting	Score	p-MRR
Threshold	$\lambda = 0.5$ (default)	28.7	+12.3
	$\lambda = 1.0$	28.3	+11.8
	$\lambda = 1.5$	28.1	+9.2
	$\lambda = 2.0$	27.8	+7.5
Temperature	$\tau = 0.1$ (default)	28.7	+12.3
	$\tau = 0.2$	28.2	+11.9
	$\tau = 0.5$	27.8	+10.4
	$\tau = 1.0$	27.1	+8.7
Candidate depth	$K = 10$	26.3	+16.1
	$K = 50$	27.2	+18.8
	$K = 100$	27.9	+19.5
	$K = 200$	27.3	+19.8
	$K = 1000$ (default)	28.7	+12.3
Query decomposition	Qwen3-4B	2.81 ± 1.20 s/q	
Post-scoring correction	Cached scores	0.4 ms/q	

Table 10: Sensitivity and efficiency with RepLLaMA on FollowIR. Each sensitivity block varies one factor while retaining the other defaults ($K=1000$, $\lambda=0.5$, and $\tau=0.1$). Candidate depth K jointly specifies the number of Q_{full} candidates retrieved, used for normalization and Huber fitting, and reranked before metric computation. Score and p-MRR are averaged over the three FollowIR datasets. Decomposition latency is measured across all FollowIR queries using Qwen3-4B on a single NVIDIA GeForce RTX 3090 GPU with temperature 0.1; we report mean $\pm$ standard deviation. The 0.4 ms measurement covers score normalization, Huber fitting, and final reranking after retrieval scores are available.

measurement. Among the reported online components, latency is dominated by the one-time query decomposition. TRIX requires no retriever training, document re-encoding, or candidate-level LLM inference; its additional LLM cost is one query-level decomposition.

C Additional Exclusion-Benchmark Results

Table 4 (main text, Section 4.3) reports the complete TRIX results across NegConstraint and ExcluIR. Table 11 examines whether its performance depends on using the per-query regression residual instead of raw exclusion similarity. The same matched comparison is applied to both benchmarks. Within each benchmark, all rows use the same frozen retriever, candidate pool, decomposition, query templates, document embeddings, and shared λ, τ . The pool contains the top 1,000 candidates for NegConstraint and the complete 90,406-document corpus for ExcluIR. The raw and residual correction rows differ only in the exclusion signal supplied to the same TRIX scoring function.

ExcluIR Evaluation Protocol. For an ExcluIR request u , let $\text{rank}_u(d) \in \{1, \dots, 90,406\}$ be the one-based rank of document d after reranking the full corpus. For an annotated desired document d^+ , its top-10 reciprocal-rank contribution is $1/\text{rank}_u(d^+)$ when $\text{rank}_u(d^+) \leq 10$ and zero otherwise; MRR averages this quantity over the evaluation instances. For each annotated desired-excluded pair (d^+, d^-) , Right Rank is one when $\text{rank}_u(d^+) < \text{rank}_u(d^-)$ and zero otherwise;

Variant	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
Full-query baseline	63.8	71.9	73.4	56.0
TRIX w/ positive view only	76.0	81.8	75.0	58.0
TRIX w/ raw exclusion score	76.6	81.8	76.0	60.0
Full TRIX	76.6	81.8	78.2	82.2

Table 11: Matched exclusion-signal ablation across exclusion benchmarks with BGE-large-en-v1.5. Within each benchmark, all variants use the same candidate pool, generated views, document representations, and scoring hyperparameters.

RR is its mean over all pairs. Equal ranks contribute zero. Full-corpus scoring guarantees that neither pair member is missing from the evaluated ranking.

Result Provenance. We reproduce DEO with BGE-large on both benchmarks. For the controlled NegConstraint comparison, DEO and TRIX start from the same BGE-large top-1,000 candidate lists and use the same benchmark split and evaluation implementation; each method retains its own query construction and scoring rule. The ExcluIR values follow the full-corpus protocol above. The SEAL ExcluIR values are taken from its source paper. The matched Base/+TRIX blocks isolate TRIX’s score correction, while the controlled DEO comparison evaluates it against another training-free method under the same NegConstraint setup.

The two benchmarks show different contributions from the exclusion signal. On NegConstraint, the positive view accounts for most of the gain over the full-query baseline: it raises MAP from 63.8 to 76.0 and nDCG@10 from 71.9 to 81.8. Raw exclusion scoring adds 0.6 MAP points and no nDCG@10 gain, while replacing it with the residual leaves both reported metrics unchanged at 76.6 and 81.8. On ExcluIR, the residual correction improves over raw exclusion scoring from 76.0 to 78.2 MRR and from 60.0 to 82.2 RR. Thus, in this matched comparison, residualization contributes most clearly to the desired-excluded pair ordering measured by ExcluIR, whereas the NegConstraint improvement is driven primarily by the positive view.

D Qualitative Analysis

The qualitative analysis draws one example from successful cases and one from unsuccessful cases on FollowIR Core17 with RepLLaMA. Each example reports the benchmark, query ID, complete request, decomposed views, candidate scores, and ranking change. The two examples illustrate how the scoring mechanism behaves under successful and unsuccessful outcomes. We define the failure pool as queries for which the final top 10 contains no gold-relevant document and MRR@10 remains zero.

E Decomposition and Reproducibility Details

The instruction decomposer is the frozen Qwen3-4B model (Yang et al. 2025). It receives the complete user request and

Outcome	Request and views	Candidate evidence	Ranking effect and interpretation
Success	Benchmark: FollowIR (Core17), QID 355 Query: "Identify documents discussing the development and application of space-borne ocean remote sensing" Exclusion: land-bound sciences, agriculture, forestry, marketing, temperature Views: Q_{full} (full query), Q_{pos} (ocean remote sensing in oceanography, seabed prospecting, marine sciences), Q_{neg} (land-bound sciences, agriculture, forestry, marketing, temperature)	The original top-1 document scores high on both s_{pos} and s_{neg} because it contains agriculture and forestry content. Residual r remains high after accounting for the shared score variation. After TRIX, an ocean-only document becomes the new top-1, while the original land-bound top-1 document drops to rank 580.	Land-bound/agriculture documents drop from top-10 to rank > 500. Ocean remote-sensing documents rise to top-5. Interpretation: TRIX uses the high residual to suppress the excluded content in this candidate set.
Failure	Benchmark: FollowIR (Core17), QID 416 Query: "What is the ongoing status of The Three Gorges Project?" Inclusion: completion date, total cost, or electrical output Exclusion: social, political, or ecological impacts and displacement Views: Q_{full} (full request), Q_{pos} (project status through completion, cost, and electrical output), Q_{neg} (social, political, and ecological impacts; displacement)	The final top-5 documents mix requested project facts with explicitly excluded impacts. Under the instruction, all five have FollowIR gold relevance 0 and belong to the benchmark's label-difference set. Their residuals range from -5.10 to -2.22, below $\lambda = 0.5$, so $h = 0$ and $g = 1$ for each document; the positive-view reward is retained without an exclusion penalty.	The original top-1 document remains at rank 1. The first gold-relevant document rises from rank 23 to 13, but none reaches the top 10, and MRR@10 remains 0. Interpretation: Their high positive-view scores raise the fitted negative-score expectation, whereas their observed negative-view scores remain below that expectation. The resulting negative residuals expose a limitation of relative residual scoring: it can miss hard-constraint violations when the excluded content does not produce negative similarity beyond that expected from the shared topic.

Table 12: One successful and one unsuccessful TRIX example drawn from FollowIR Core17 with RepLLaMA. Each example reports the operational views, score-space evidence, and ranking change under the complete scoring rule.

returns structured affirmative and exclusion content in one generation. The full view Q_{full} is the original request itself, used unchanged for candidate generation; Q_{pos} retains the topic and affirmative conditions while removing exclusion content, and Q_{neg} contains the excluded content, or [NONE] when no exclusion is present. We use the fixed prompt with temperature 0.1, top- $p = 0.9$, at most 512 new tokens, and thinking disabled. Malformed or missing fields are mapped to the corresponding empty view. The system prompt is:

You are an expert IR query optimizer for Dense Vector Models. Decouple "Original Query" and "Instruction" into Q_{plus} and Q_{minus} .

Q_{plus} rules: (1) Fluent natural language phrase, not a keyword list. (2) Semantic topic only; remove format noise ("articles", "documents", "reports"). (3) Remove conversational connectors ("such as", "including"). (4) Sharp, coherent.

Q_{minus} rules: (1) Extract only critical high-level exclusion concepts. (2) Concise natural phrasing. (3) If no exclusion, output "[NONE]".

No-negation rule: DENSE RETRIEVERS IGNORE LOGIC WORDS. NEVER use negation in Q_{minus} ("no", "not", "non-", "without"). Convert negations to AFFIRMATIVE entities:
 BAD: "not colleges in Manhattan"
 -> GOOD: "colleges in Manhattan, such as Columbia University"
 Output ONLY exact entities to exclude.

Output JSON:

```
{
  "Reasoning_Steps": "1. Core topic.
  2. Filter noise. 3. Exclusions (negations->affirm).",
  "Q_plus": "Fluent natural language query",
  "Q_minus": "Core exclusions or [NONE]"
}
```

The user prompt template is:

Decouple into Q_{plus} and Q_{minus} :
 Query: "{query}"
 Instruction: "{instruction}"

For every backbone, including RepLLaMA, all document representations are computed once with that retriever's na-

Benchmark	N	Parse	Q_+ ok	Q_- ok	[NONE] Q_-
FollowIR	208	100.0%	100.0%	100.0%	38.0%
NegConstraint	198	100.0%	100.0%	100.0%	0.5%
ExcluIR	3,452	100.0%	100.0%	100.0%	0.4%

Table 13: Full structural validity of decomposed records across benchmarks. “ Q_+ ok” and “ Q_- ok” denote field-existence rates; “[NONE] Q_- ” denotes an explicit [NONE] value in the single- Q_- output. All fields parse successfully; no malformed or missing-field fallback is triggered.

tive document template, and all three query views use its native query template. Matched Base/+TRIX variants reuse the same candidates, templates, and document representations. The $K_{\text{retrieve}} = K_{\text{fit}} = K_{\text{rerank}} = 1000$ convention in Section 4.1 applies to the reported FollowIR, NegConstraint, and BEIR results. ExcluIR uses the complete 90,406-document corpus as $\mathcal{C}_u$ for score normalization, Huber fitting, and reranking.

Numerical and Scoring Constants. We set the Huber transition point to $\delta = 1.35$ in standardized score units. The constant $\epsilon = 10^{-6}$ serves both as the median/MAD denominator floor and as the low-dispersion threshold: when $\text{MAD}(S_{\text{pos}}) < \epsilon$, the fitted slope is set to zero. The regression is fit independently on each candidate set and uses no relevance labels. Using RepLLaMA on the FollowIR training corpus, we search $\lambda \in [0.05, 5.0]$ and $\tau \in [0.01, 1.0]$ and select $(\lambda, \tau) = (0.5, 0.1)$ based on aggregate Score and p-MRR. We then freeze this pair for all evaluation datasets and retrievers, without per-evaluation-dataset or per-retriever tuning. Because r is centered and scaled by its within-query MAD, $\lambda = 0.5$ activates the correction half a residual-MAD unit above the median fitted deviation. With $\tau = 0.1$, each additional 0.1 active residual unit multiplies the affirmative reward by $\exp(-1)$. The full-query score, attenuated affirmative reward, and residual penalty use unit coefficients; no query-specific scoring weights are introduced.

F Decomposition Quality Audit

We audit the structural and semantic quality of the Qwen3-4B decomposer output on all three benchmarks used in the paper.

F.1 Full Structural Validity

Table 13 reports automated checks on every decomposed record. All 3,858 records parse successfully as valid JSON. The Q_+ field is present and non-trivial in every record. The Q_- field is present in every record. The observed [NONE] rates are 0.5% on NegConstraint and 0.4% on ExcluIR. On FollowIR, 38.0% of queries yield $Q_- = [\text{NONE}]$. These checks establish field-level validity; the semantic correctness of [NONE] assignments is assessed separately in Section F.2. No empty-string Q_+ or malformed-field fallback is triggered in any benchmark.

Criterion	FollowIR			NegConstr.			ExcluIR	
	Y	N	U	Y	N	U	Y	N
C1: Q_{pos} topic	30	0	0	30	0	0	30	0
C2: Q_{pos} affirmative	30	0	0	30	0	0	30	0
C3: Q_{pos} no exclusion leak	30	0	0	30	0	0	30	0
C4: Q_{neg} covers excl.	18	0	2	29	0	0	20	0
C5: Q_{neg} no hallucination	18	0	2	29	0	0	20	0
C6: [NONE] correct	10	0	0	1	0	0	8	2
C7: Structure correct	28	0	2	30	0	0	27	3

Table 14: Human semantic audit results ($T=30$ per benchmark, 90 total). C4/C5 are evaluated only on records with effective Q_- (FollowIR $n=20$, NegConstraint $n=29$, ExcluIR $n=20$); C6 only on records with $Q_-=[\text{NONE}]$ ($n=10$, 1, 10 respectively). C3 records whether Q_{pos} avoids retaining exclusion content; all 90 audited records pass this criterion.

F.2 Human Semantic Audit

We draw a stratified random sample of $T=30$ records per benchmark (90 total), oversampling records with effective Q_- within each stratum. One author independently labels each record on seven binary/ternary criteria:

1. Q_{pos} preserves the original topic.
2. Q_{pos} preserves all affirmative (inclusion) conditions.
3. Q_{pos} does *not* erroneously retain exclusion content.
4. Q_{neg} covers the exclusion condition(s) present in the request.
5. Q_{neg} does *not* introduce entities or scope absent from the original request (hallucination).
6. Where $Q_-=[\text{NONE}]$, the [NONE] assignment is correct.
7. Scope, time, geographic, entity, and exception structures are correctly handled.

Labels are Y/N/U (yes/no/unclear), with NA for criterion 6 when $Q_- \neq [\text{NONE}]$. Table 14 reports per-benchmark pass rates.

Key findings. Q_{pos} preserves topic and affirmative conditions in all 90 records (100%); no record leaks exclusion content into Q_{pos} . Q_{neg} correctly covers the exclusion condition in 97.1% of effective- Q_- records (67/69); the two unclear FollowIR cases involve negation phrasing within Q_- . For hallucination, 67/69 effective- Q_- records introduce no entities or scope absent from the original request, and the two remaining FollowIR cases are unclear. The [NONE] assignment is correct in 90.5% of cases (19/21); the 2 failures are both on ExcluIR, where requests containing exclusion cues are incorrectly marked [NONE]. Scope, time, and entity structures are correctly handled in 94.4% of records (85/90; 2 unclear on FollowIR and 3 failures on ExcluIR).

F.3 List-vs-Single Exclusion Decomposition

The Qwen3-4B decomposer can emit Q_- either as a single string or as a list of independent exclusion units $Q_- = [Q_{-1}, \dots, Q_{-M}]$. When $M > 1$, the per-unit residuals $r_j(d)$ must be aggregated before the monotone composition (Eq. 7). We compare three strategies:

Q- Strategy	Core17	Robust04	News21	Avg
<i>p-MRR (higher is better)</i>				
Single	0.162	0.141	0.068	0.124
List, max	0.263	0.254	0.175	0.231
List, mean	0.212	0.290	0.193	0.232
<i>changed MAP@1000 (higher is better)</i>				
Single	0.259	0.282	0.321	0.287
List, max	0.246	0.268	0.271	0.262
List, mean	0.269	0.246	0.255	0.257

Table 15: List-vs-single Q_- decomposition with residual aggregation strategies on FollowIR (RepLLaMA backbone, TRIX full). p-MRR is per-task; changed MAP@1000 uses the task-specific definition from Section 4.1.

- **Max-residual:** $r_{\text{agg}} = \max_j r_j(d)$. Takes the most aggressive exclusion signal.
- **Mean-residual:** $r_{\text{agg}} = \frac{1}{M} \sum_j r_j(d)$. Averages exclusion evidence, avoids over-penalization.
- **Single Q-:** The decomposer outputs one merged exclusion string; no aggregation is needed.

Table 15 reports RepLLaMA + TRIX (full) on FollowIR. The list-based decomposition with max-residual pooling improves avg. p-MRR by +0.107 over the single-Q baseline, with gains on all three datasets (Core17 +0.101, Robust04 +0.113, News21 +0.107). Mean-residual pooling yields a comparable gain (+0.108), driven primarily by Robust04 (+0.149). Both list variants reduce changed MAP@1000 (−0.025 and −0.030), indicating that the more aggressive multi-exclusion penalty can occasionally over-suppress relevant documents. Because the p-MRR improvement is accompanied by a consistent MAP reduction and the single-Q-design is simpler, we adopt the single-Q- formulation as the default in the paper.

G General Retrieval Evaluation

To assess whether TRIX affects general retrieval effectiveness, we compare the frozen BGE-large-en-v1.5 baseline with its TRIX-reranked results on eight BEIR benchmarks.

Table 16 reports nDCG@10 and MAP@100 for each dataset. TRIX maintains general retrieval effectiveness overall: five datasets improve in nDCG@10 (+0.05 to +2.26 points), while three show small decreases (−0.01 to −1.03 points). The average changes are +0.21 nDCG@10 and −0.05 MAP@100.

Across the eight datasets, TRIX preserves average general retrieval performance, improving nDCG@10 by 0.21 points while MAP@100 changes by only −0.05 points. NQ and HotpotQA show nDCG@10 changes of at most 0.09 points.

Dataset	nDCG@10 ↑	Δ	MAP@100 ↑	Δ
trec-covid	71.40	+2.26	8.70	+0.20
fiqa	42.45	+0.76	36.38	+0.66
hotpotqa	73.29	+0.09	66.06	+0.03
nq	51.03	+0.05	44.02	+0.04
nfcopus	36.53	+0.05	17.68	+0.17
scifact	74.12	−0.01	69.83	−0.07
arguana	45.87	−0.44	32.80	−0.26
quora	87.17	−1.03	83.92	−1.18
Average	60.23	+0.21	44.92	−0.05

Table 16: BEIR evaluation with BGE-large and TRIX. Values are percentages ($\times 100$). Δ shows absolute change relative to the BGE-large baseline.